

Numerical Method

CSIT — 3rd Semester — 2081 — Answer Sheet
Subject Code: CSC212

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. What are inherent errors? Derive the Newton Raphson method for solving non-linear equation and using this method solve $x^2 - 5x + 6 = 0$, calculate upto 3 decimal places.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What are the limitations of direct methods for solving a system of linear equations? How Gauss Seidel method differs from Jacobi iteration? Solve the following system of linear equation using Jacobi iteration method. $2x - 7y - 10z = 175$, $x + y + 3z = 14$, $x + 10y + 9z = 7$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Write an algorithm and program to implement Lagrange interpolation method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. Consider the following data points estimate the $f(0.6)$ using Newton's interpolation formula.
x: 0.10, 0.20, 0.30, 0.40, 0.50
f(x): 2.683, 0.043, 3.383, 6.93, 9.7

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is regression analysis? Fit a second order polynomial for the following data values.
x: 24, 68, 101, 42, 02, 42, 62, 8

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is numerical differentiation? The table below gives the values of distance travelled by a vehicle at various time interval, estimate the velocity and acceleration at $x=4$.
Time (x): 1, 2, 4, 8, 10
Distance (y): 0, 15, 21, 27

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is application of numerical integration? Find the value of integration for $\int_0^2 e^x/x \, dx$ using Simpson's 3/8 rule with $n=6$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Solve the following system of linear equations using Gauss-Jordan elimination method.
 $x+2y-3z=4$
 $2x+4y-6z=8$
 $x-2y+5z=4$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Given the data points below

x	f(x)
1.03	0.04
1.54	0.59
2.05	1.04

Find cubic spline which belongs to $1 \leq x \leq 3$ and estimate $f(2)$ using cubic splines.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is differential equation? Differentiate between ODE and PDE with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Solve $dy/dx = x/y$, $y(0)=1$, at $x=0.4$ using Runge-Kutta's 4th order method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Solve the Poisson equation $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 64xy$, $0 \leq x \leq 1$, $0 \leq y \leq 1$ with boundary conditions: $u(0,y)=0$, $u(x,0)=0$, $u(1,y)=150$, $u(x,1)=150$ and $h=1/3$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.